



Inspiring Excellence

CSE422: Artificial Intelligence Project
Name: Customer Churn Prediction
Group: 03

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Submission Date: January 06, 2026

1. Introduction

The primary goal of this project is to build a machine learning system that can predict Customer Churn. Churn occurs when a customer stops doing business with an entity. Predicting this allows a company to proactively offer incentives to keep the customer. In a competitive market, acquiring new customers is more expensive than retaining existing ones. Our problem is to identify which customers are likely to leave based on their demographic (Age, Gender) and financial (Income, Credit Score) profiles. By using Artificial Intelligence, we can move from "guessing" why people leave to "predicting" it with data. This project explores how different algorithms from simple linear models to complex neural networks handle a dataset where patterns are not immediately obvious.

2. Dataset Description

- **Number of Data Points:** 1,500 rows.
- **Number of Features:** 9 columns (8 independent variables + 1 target variable).
- **Feature Types:**
 - **Numerical:** Age, Income, Credit_Score, Num_Purchases, Membership_Years.
 - **Categorical:** Gender, Marital_Status, Device_Used.
 - **Target:** Churn (Binary: 0 or 1)

We split the data features based on the numerical data and categorical data.

3. Data Analysis (EDA)

Exploratory Data Analysis (EDA) is the "discovery" phase where we summarize the main characteristics of the dataset using statistical and visual methods to get inference about the feature.

A. Statistical Profiling

- **Dimensions:** We confirmed the dataset has **1,500 rows and 9 columns**.
- **Missing Values:** We detected **75 null entries** in Age, Income, and Credit Score using `df.isna().sum()`.
- **Unique Values:** We categorized features into **Continuous** (Income) and **Discrete** (Num_Purchases, Churn) using `nunique()`.

B. Symmetry and Skewness

We checked the **Skewness** of our numerical features to see if the data was balanced around the mean.

- **Finding:** While all the features were relatively symmetric, we observed that **Num_Purchases was Right-Skewed** (Positively Skewed).
- **Meaning:** This means most customers make a small number of purchases, with only a few "power users" making a very high number of purchases, creating a "long tail" on the right side of the graph.

C. Outlier Detection (Boxplots)

We used a loop to create boxplots for every numerical feature.

- **Finding:** The boxplots clearly showed **Outliers** (points outside the "whiskers"). We spotted impossible data: negative Income and Credit_Score values above 850.
- **Action:** These plots proved that the data needed "cleaning" before it could be used.

D. Correlation Analysis

- **Heatmap:** We checked how features relate to each other.
- **Result:** We found **very low correlation** between demographics (Age, Income) and Churn. This told us that predicting churn would be a difficult task with these specific features.

4. Data Pre-processing

Data pre-processing is where we fix the problems found in the Analysis phase to make the data "machine-readable."

Step 1: Outlier Removal (Cleaning)

- **Problem:** We found "dirty" data like negative income and credit scores above 850.
- **Solution:** We filtered the dataset to remove these impossible values to prevent the models from learning incorrect patterns.

Step 2: Missing Value Imputation

- **Problem:** 75 missing values in Age, Income, Credit Score and Gender column.
- **Solution:** We used the **SimpleImputer**.
 - **Numerical:** Used **Median** (Middle value). We chose Median over Mean because the median is not affected by the **Skewness** we found in features like Num_Purchases.
 - **Categorical:** Used **Most Frequent** (Mode) to fill missing Gender entries.

Step 3: Categorical Encoding

- **Problem:** Computers only understand numbers, not words like "Mobile" or "Married."
- **Solution:** We applied **One-Hot Encoding**. This turns one column into several binary (0 or 1) columns. We used drop_first=True to keep the data mathematically efficient.

Step 4: Handling Skewness & Scaling

- **Problem:** Features have different units (years, dollars) and different distributions (some skewed, some normal).
- **Solution:** We applied **Standardization** (StandardScaler).
- **Result:** This process centers the data and reduces the impact of **Skewness**, making it easier for algorithms to find the "global minimum" during training.

Step 5: Train-Test Splitting

We divided the data into **Training (80%)** and **Testing (20%)**. This allows us to train the model on one set and then verify its accuracy on "unseen" data, ensuring it hasn't just memorized the answers.

5. Model Training

As per the guideline, we applied one Neural Network, two classification models, and one clustering model.

I. Logistic Regression

This model predicts the probability of a class using the **Sigmoid Function**. It takes a linear combination of features and squashes the result into a range between 0 and 1.

II. Random Forest Classifier

This is an **Ensemble** model. Instead of one decision tree, it builds 500 trees using random subsets of data and features.

- **Logic:** It uses "Majority Voting". If among 100 trees 70 trees say the customer will stay and 30 say they will leave, the final output is "Stay."

III. Multi-Layer Perceptron (MLP - Neural Network)

It consists of an input layer, hidden layers of neurons, and an output layer.

- **Logic:** Each neuron has a **Weight** (w) and a **Bias** (b). It uses an activation function (like ReLU) to decide if a signal is strong enough to pass to the next layer.

IV. K-Means Clustering (Unsupervised)

We treated the problem as unsupervised to see if customers naturally group together.

- **Logic:** It places K centroids randomly and assigns each data point to the nearest one using **Euclidean Distance**.

6. Model Evaluation

We evaluated the models using **Accuracy Score** on a 20% test split.

Model	Accuracy
K-Means	56.67%
Random Forest	50.00%
Logistic Regression	51.00%
MLP (Neural Network)	49.33%

7. Conclusion

Among all the models **K-Means Clustering** model performed best, though only by a small margin (56.67%). It performed better because it ignored the "noisy" churn labels and instead identified the real, natural groupings already present within the feature data itself.

Probable Reasons for Results

1. **Independent Features:** The demographic data provided (Age, Income) does not seem to be the actual reason why customers leave in this specific dataset.

2. **Data Noise:** The lack of correlation between features suggests the data might be random or missing the most important "behavioral" features (e.g., how many times they called customer support).
3. **Neural Network Complexity:** The MLP performed worst because Neural Networks usually require much larger datasets and stronger patterns to beat simpler models like Logistic Regression.